

LADYSMITH, South Africa

WMO: 684790

Lat: 28.5756S

Lon: 29.7503E

Elev: 3507

StdP: 12.93

Time Zone: 2.00 (E02)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.0	36.7	11.1	11.0	61.1	16.5	14.4	59.6	24.4	58.1	20.3	65.0	2.7	290	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.8	92.9	64.1	90.3	64.1	87.6	64.2	70.9	82.0	69.6	80.2	68.5	79.0	9.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	117.3	74.4	66.6	111.7	73.4	65.5	107.5	72.7	37.2	82.8	36.0	80.0	35.0	79.1	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	16.4	13.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.3	72.6	72.8	70.4	64.5	60.1	55.4	54.9	59.4	65.2	67.4	69.7	72.2		
	DBStd	8.64	4.64	4.62	4.84	5.05	5.15	4.92	5.27	6.80	8.01	7.40	6.86	5.98		
	HDD50	38	0	0	0	0	1	8	17	8	4	0	0	0		
	HDD65	1265	5	4	13	69	167	290	315	196	95	63	35	13		
	CDD50	5641	701	639	633	434	314	171	168	301	460	541	591	688		
	CDD65	1389	241	223	181	53	15	2	1	24	100	138	176	235		
	CDH74	10559	1643	1385	1113	323	169	34	26	276	1034	1268	1410	1878		
	CDH80	3721	555	447	339	54	24	3	0	56	403	514	555	771		
(14) Wind	WSAvg	4.9	5.1	4.6	4.2	3.9	3.0	4.6	3.4	4.7	6.2	6.5	6.7	6.3		
(15) Precipitation	PrecAvg	29.0	5.5	4.2	3.6	1.8	0.6	0.3	0.3	0.9	1.3	2.9	3.5	4.3		
	PrecMax	45.6	12.2	9.3	7.5	8.9	3.3	2.3	3.0	3.9	10.1	7.0	7.0	9.6		
	PrecMin	15.7	1.0	0.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.2	0.7		
	PrecStd	7.5	2.6	2.3	1.8	1.7	0.8	0.5	0.6	1.1	1.9	1.9	1.7	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.2	93.0	91.4	85.6	83.3	77.4	76.9	86.6	93.6	95.3	95.5	97.5		
		MCWB	65.8	66.8	66.5	60.8	56.6	51.9	51.1	57.9	59.7	63.0	64.5	65.2		
	2%	DB	89.6	89.0	86.9	81.1	78.7	73.4	73.6	80.8	88.8	90.3	90.5	91.7		
		MCWB	66.5	67.6	65.7	60.3	56.4	50.3	50.4	54.2	58.1	61.0	63.0	65.4		
	5%	DB	86.2	85.5	83.8	77.7	74.9	70.0	70.6	76.8	84.3	85.9	86.4	87.9		
		MCWB	66.8	67.1	65.2	60.0	55.2	50.4	50.2	53.0	57.0	60.9	63.3	65.9		
	10%	DB	82.7	82.2	80.4	74.3	71.0	66.6	67.2	72.9	79.5	81.3	82.2	84.1		
		MCWB	66.7	66.9	64.9	59.4	54.8	49.9	49.1	52.6	56.3	60.4	63.3	66.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.8	72.4	70.7	65.8	60.7	56.4	55.5	61.3	63.5	68.1	70.0	72.0		
		MCDB	82.3	84.3	82.8	76.8	74.7	68.1	67.5	76.0	81.0	83.3	82.6	83.7		
	2%	WB	70.7	70.6	68.8	63.9	58.9	53.9	53.8	58.7	61.6	65.4	68.0	69.7		
		MCDB	80.7	81.4	79.5	73.7	70.3	66.9	64.2	72.2	78.5	79.4	80.6	81.2		
	5%	WB	69.2	69.4	67.4	62.6	57.5	52.6	52.2	56.6	60.1	63.9	66.4	68.3		
		MCDB	79.4	79.3	77.4	71.7	69.7	64.3	64.1	70.9	76.1	77.6	78.0	79.4		
	10%	WB	68.0	68.3	66.3	61.0	56.0	51.3	50.9	54.5	58.7	62.6	65.2	67.2		
		MCDB	78.1	78.1	75.8	69.8	68.1	62.4	63.1	68.0	73.9	75.1	76.2	78.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.8	19.3	19.5	20.9	24.6	25.4	26.4	24.8	24.4	22.1	21.5	21.0		
		MCDBR	25.8	24.1	25.1	26.3	29.0	29.3	30.0	30.0	32.0	31.3	29.4	27.8		
	5% WB	MCWBR	7.6	6.9	7.3	8.1	11.1	12.9	13.2	11.3	10.1	8.4	7.4	7.3		
		MCWBR	21.3	21.4	21.3	21.2	25.1	24.1	25.2	25.9	26.7	26.0	23.4	23.4		
(40) Clear-Sky Solar Irradiance	taub		0.391	0.382	0.371	0.335	0.299	0.283	0.288	0.349	0.393	0.427	0.388	0.391		
	taud		2.353	2.382	2.390	2.448	2.512	2.525	2.496	2.332	2.234	2.189	2.329	2.342		
	Ebn at Noon		302	298	292	287	285	284	286	278	280	282	301	303		
	Edh at Noon		42	40	37	32	27	25	27	35	43	48	43	43		
(44) All-Sky Solar Radiation	RadAvg		2024	1936	1708	1427	1254	1137	1221	1432	1680	1767	1988	2070		
	RadStd		148	161	109	91	65	62	60	84	95	139	137	107		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	+0.59	+1.03	N/A	N/A	-0.61	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page